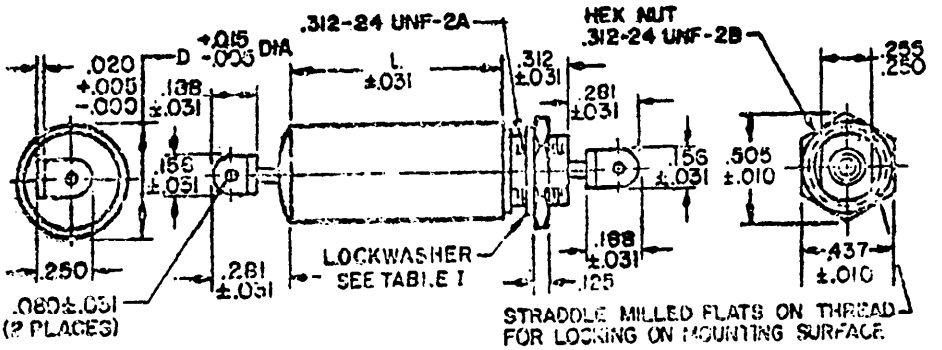
MIL-C-83439/6B(USAF)
08 August 1977
SUPERSEDING
MIL-C-83439/6A(USAF)
20 May 1976

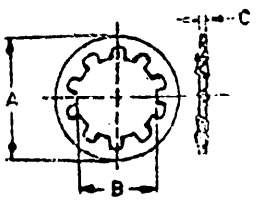
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, FEED THROUGH, ELECTROMAGNETIC INTERFERENCE SUPPRESSION, (HERMETICALLY SEALED IN METAL CASES)

This specification sheet is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-83439.







CIRCUIT CONFIGURATION

Table I. Hardware dimensions.

Case Diameter D	Lockwasher 1/		
	A	B	C
.400 and .500	.430 ±.005	.318 ±.004	.023 ±.003

1/ For .562 and larger use MS35333-75

INCHES	MM	INCHES	MM
.003	.08	.250	6.35
.004	.10	.255	6.48
.005	.13	.281	7.14
.010	.25	.312	7.92
.015	.38	.318	8.08
.020	.51	.400	10.16
.023	.58	.430	10.92
.031	.79	.437	11.10
.080	2.03	.500	12.70
.125	3.18	.505	12.83
.156	3.96	.562	14.27
.188	4.78		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Circuit diagram is for information only.
4. Mounting hardware (lockwasher and hex-nut) shall be supplied with capacitor.
5. Use of shoulder is optional.
6. Unless otherwise specified, tolerance is ±.005 (.13 mm).

FIGURE 1. Case dimensions and circuit configurations.

FSC 5910

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TABLE II. Dimensions and electrical characteristics.

Capacitance value	Rated voltage 1/				Case dimensions		Part numbers
µF	DC volts	400 Hz AC volts	1200 Hz AC volts	2000 Hz AC volts	+.015 D-.005	L ±.031	M83439/06-
.18	50	35	25	15	0.400	0.687	0001
.22	↓	↓	↓	↓	↓	↓	0002
.27							0003
.33							0004
.39							0005
.47							0006
.56							0007
.68							0008
.82						↓	0009
1.0						0.687	0010
1.2						0.937	0011
1.5						↓	0012
1.8							0013
2.2		↓	↓	↓		0.937	0014
2.7		35	25	15	↓	1.187	0015
3.3		30	20	12	0.400	↓	0016
3.9		↓	↓	↓	0.500		0017
4.7					0.500		0018
5.6		30	20	12	0.500	↓	0019
6.8		25	15	10	0.562	1.187	0020
8.2		↓	↓	↓	↓	1.375	0021
10.0		↓	↓	↓		1.625	0022
12.0		25	15	10	0.562	↓	0023
15.0		20	12	8	0.670		0024
18.0		↓	↓	↓	0.670	1.625	0025
22.0	↓	↓	↓	↓	0.670	1.875	0026
27.0	50	20	12	8	0.750	1.875	0027
.15	75	50	30	20	0.400	0.687	0028
.18	↓	↓	↓	↓	↓	↓	0029
.22							0030
.27							0031
.33							0032
.39						↓	0033
.47						0.687	0034
.56						0.687	0035
.68						0.687	0036
.82						0.937	0037
1.0						0.937	0038
1.2						0.937	0039
1.5						1.187	0040
1.8						↓	0041
2.2					0.400		0042
2.7					0.500		0043
3.3					0.500		0044
3.9					0.562	↓	0045
4.7		↓	↓	↓	↓	1.187	0046
5.6		50	30	20		1.375	0047
6.8		40	25	15	0.562	1.625	0048
8.2		↓	↓	↓	0.670	1.625	0049
10.0		↓	↓	↓	0.670	1.625	0050
12.0		40	25	15	0.670	1.875	0051
15.0	↓	30	18	12	0.750	1.875	0052
18.0	75	30	18	12	0.750	1.875	0053

See footnote at end of table.

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TABLE II. Dimensions and electrical characteristics.-Continued.

Capacitance value	Rated voltage 1/				Case dimensions		Part numbers
μF	DC volts	400 Hz AC volts	1200 Hz AC volts	2000 Hz AC volts	+.015 D-.005	L ±.031	M83439/06-
.068	100	65	40	30	0.400	0.687	0054
.082	↓	↓	↓	↓	↓	↓	0055
.10							0056
.12							0057
.15							0058
.18							0059
.22							0060
.27						0.687	0061
.33						0.687	0062
.39						0.687	0063
.47						0.937	0064
.56						0.937	0065
.68						0.937	0066
.82						1.187	0067
1.0						↓	0068
1.2					0.400		0069
1.5					0.500		0070
1.8					0.500		0071
2.2					0.562		0072
2.7					↓	1.187	0073
3.3						1.375	0074
3.9						1.625	0075
4.7		↓	↓	↓	↓	↓	0076
5.6		65	40	30	0.562		0077
6.8		45	30	20	0.670		0078
8.2		↓	↓	↓	0.670	1.625	0079
10.0					0.670	1.875	0080
12.0	100	45	30	20	0.750	1.875	0081
.047	150	90	55	40	0.400	0.687	0082
.056	↓	↓	↓	↓	↓	↓	0083
.068							0084
.082							0085
.10							0086
.12							0087
.15							0088
.18							0089
.22						0.687	0090
.27						0.937	0091
.33						↓	0092
.39						0.937	0093
.47						1.187	0094
.56						↓	0095
.68							0096
.82					0.400		0097
1.0					0.500		0098
1.2					0.500		0099
1.5					0.562		0100
1.8					↓	1.187	0101
2.2						1.375	0102
2.7					0.562	1.625	0103
3.3		90	55	40	0.670	1.625	0104
3.9		80	45	35	0.670	1.625	0105
4.7		↓	↓	↓	0.670	1.875	0106
5.6					0.750	1.875	0107
6.8	150	80	45	35	0.750	1.875	0108

See footnote at end of table.

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TABLE II. Dimensions and electrical characteristics.-Continued.

Capacitance value	Rated voltage 1/				Case dimensions		Part numbers
µF	DC volts	400 Hz AC volts	1200 Hz AC volts	2000 Hz AC volts	+.015 D-.005	L ±.031	M83439/06-
.010	200	120	80	60	0.400	0.687	0109
.012							0110
.015							0111
.018							0112
.022							0113
.027							0114
.033							0115
.039							0116
.047							0117
.056							0118
.068							0119
.082							0120
.10							0121
.12							0122
.15							0123
.18						0.687	0124
.22						0.937	0125
.27						0.937	0126
.33						0.937	0127
.39						1.187	0128
.47							0129
.56					0.400		0130
.68					0.500		0131
.82					0.500		0132
1.0					0.562		0133
1.2						1.187	0134
1.5						1.375	0135
1.8		120	80	60	0.562	1.625	0136
2.2		95	65	50	0.670	1.625	0137
2.7					0.670	1.625	0138
3.3					0.670	1.875	0139
3.9					0.750	1.875	0140
4.7	200	95	65	50	0.750	1.875	0141
.039	300	200	115	90	0.400	0.687	0142
.047						0.687	0143
.056						0.750	0144
.068						0.750	0145
.082						0.875	0146
.10						0.875	0147
.12						1.000	0148
.15						1.000	0149
.18					0.400	1.187	0150
.22					0.500		0151
.27					0.500		0152
.33					0.500		0153
.39					0.562	1.187	0154
.47						1.375	0155
.56						1.375	0156
.68					0.562	1.625	0157
.82					0.670	1.625	0158
1.0					0.670	1.625	0159
1.2					0.670	1.875	0160
1.5	300	200	115	90	0.750	1.875	0161

See footnote at end of table.

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TABLE II. Dimensions and electrical characteristics.-Continued.

Capacitance value	Rated voltage 1/				Case dimensions		Part numbers
µF	DC volts	400 Hz AC volts	1200 Hz AC volts	2000 Hz AC volts	+.015 D-.005	L ±.031	M83439/06-
.010	400	240	140	110	0.400	0.687	0162
.012							0163
.015							0164
.018							0165
.022							0166
.027							0167
.033						0.687	0168
.039						0.750	0169
.047						0.750	0170
.056						0.875	0171
.068						0.875	0172
.082						1.000	0173
.10						1.000	0174
.12					0.400	1.187	0175
.15					0.500		0176
.18					0.500		0177
.22					0.500		0178
.27					0.562	1.187	0179
.33						1.375	0180
.39						1.625	0181
.47					0.562		0182
.56					0.670		0183
.68					0.670	1.625	0184
.82					0.670	1.875	0185
1.0	400	240	140	110	0.750	1.875	0186

1/ AC voltage ratings are for design information only.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table II.

Material:
Case and terminals: Metal.
Finish: Corrosion-resistant.
Dielectric: Metallized polycarbonate.

Temperature range: -55° to +125°C. (DC ratings).
-55° to +105°C. (AC ratings).

Rated voltage: See table II.

Capacitance: See table II.
Test frequency: 1 kHz.

Capacitance tolerance: ±10 percent.

Terminal strength: Method 211 of MIL-STD-202, test-condition letter A.
Pull: 3 pounds, $^{+3}_{-0}$ ounces.

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Insulation resistance:
In megohms:
At +25° ±3°C (need not exceed) - - - - - 100,000
At +125° +4, -0°C (need not exceed)- - - 25,000
In megohms x microfarads (minimum):
At +25° ±3°C - - - - - - - - - - - - - - - 50,000
At 125° +4, -0°C - - - - - - - - - - - - - 2,000
Test potential: DC rated voltage.

Dissipation factor: 0.3 percent maximum at 25°C measured at 1,000 ±100 Hz (for capacitors having a nominal capacitance of 1 µF or less) or 100 ±10 Hz (for capacitors having a nominal capacitance greater than 1 µF).

Insertion loss (abbreviated): MIL-STD-220. See figure 2. Measurements shall be made at the following frequencies for the capacitance values specified:

Nominal capacitance (µF)	Frequency (MHz)
.010 thru .082	400 and 1000
.10 thru .18	400 and 200
.22 thru .39	400 and 100
.47 thru .82	400 and 80
1.0 thru 2.2	400 and 50
2.7 thru 10.0	400 and 10
12.0 thru 18.0	400 and 2
22.0 thru 27.0	400 and 1

Seal: Method 112 of MIL-STD-202, test condition letter - B, or in accordance with MIL-C-83439 (alternative test).

Voltage drop: 0.15 volt, maximum.

Rated current: 15 amperes.

DC resistance: 0.01 ohm, maximum.

Thermal shock: Test-condition letter D (Step 3, 125°C).

Immersion: Test-condition letter B.
Capacitance change: Not more than ±20 percent of initial measurement.

Vibration: Test-condition letter A.

Moisture resistance:
Polarization voltage: 100 Vdc.
Final measurements:
Capacitance change: Shall be not more than ±2 percent.
Dissipation factor: Shall not exceed 0.6 percent.
Insulation resistance: Shall be not less than 60 percent of initial value specified.

Life at +85°C: Not applicable.

Life at +125°C: Method 108 of MIL-STD-202, test-condition B.
Operating conditions: 1.4 times dc rated voltage (sample lot shall be 24 units).
Measurements after test:
Insulation resistance at 25°C: Shall be not less than 5,000 megohm microfarads or not less than 25,000 megohms.
Capacitance: Shall change not more than 2 percent of value specified in table II.
Dissipation factor: Shall be not more than 0.3 percent at 25°C.

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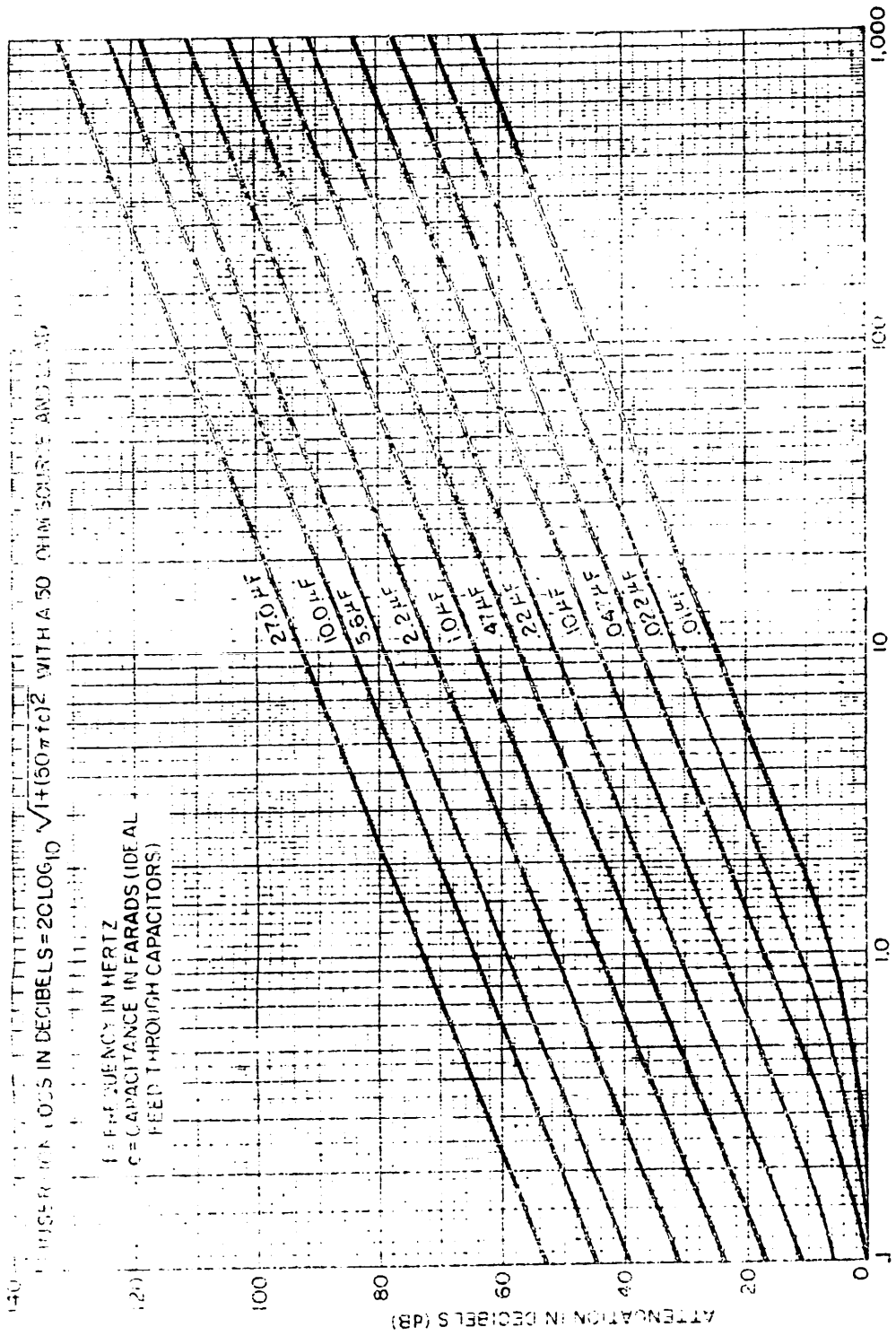


FIGURE 2. Insertion-loss versus frequency for an ideal capacitor.

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NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 85, 99
DLA - ES

Preparing activity:
Air Force - 85

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